

4. 计算下列 n 阶行列式:

$$(1) \begin{vmatrix} a_1 & a_2 & a_3 & \cdots & a_n \\ b_2 & 1 & 0 & \cdots & 0 \\ b_3 & 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ b_n & 0 & 0 & \cdots & 1 \end{vmatrix}; \quad (2) \begin{vmatrix} a_1+b_1 & a_1+b_2 & \cdots & a_1+b_n \\ a_2+b_1 & a_2+b_2 & \cdots & a_2+b_n \\ \vdots & \vdots & \ddots & \vdots \\ a_n+b_1 & a_n+b_2 & \cdots & a_n+b_n \end{vmatrix};$$

$$(3) \begin{vmatrix} a_{11}+a_{1n} & a_{11}+a_{12} & a_{12}+a_{13} & \cdots & a_{1,n-1}+a_{1n} \\ a_{21}+a_{2n} & a_{21}+a_{22} & a_{22}+a_{23} & \cdots & a_{2,n-1}+a_{2n} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ a_{n1}+a_{nn} & a_{n1}+a_{n2} & a_{n2}+a_{n3} & \cdots & a_{n,n-1}+a_{nn} \end{vmatrix};$$

$$(4) \begin{vmatrix} 1 & 2 & 3 & \cdots & n-2 & n-1 & n \\ 2 & 3 & 4 & \cdots & n-1 & n & n \\ 3 & 4 & 5 & \cdots & n & n & n \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots & \vdots \\ n & n & n & \cdots & n & n & n \end{vmatrix}.$$

4. (1) 第一行选择 a_i , 则第 i 行只能选择 b_i , 其余均为 1

$$\text{原式} = a_1 + \sum_{i=2}^n (-1)^{i+1} \cdot a_i b_i. \quad a_i - \sum_{i=2}^n a_i b_i \text{ (逆序数恒为奇)} \quad 123 \dots n \text{ 换位一次}$$

(2) 由于两行/列成比例值为 0, 故原式 = 0

$$n=2 \text{ 时原式} = \begin{vmatrix} a_1 & b_2 \\ a_2 & b_2 \end{vmatrix} + \begin{vmatrix} b_1 & a_1 \\ b_1 & a_2 \end{vmatrix} = a_1 b_2 - a_2 b_2 + a_2 b_1 - a_1 b_1 \\ = (a_1 - a_2) \cdot (b_1 - b_2)$$

$$n=1 \text{ 时} = a_1 + b_1$$

(3) 对列作操作 ① - ② + ③ - ④ + ... + ... - ⑦ + ⑧

① 若 n 为偶数, $\Rightarrow$ ① + ③ - ② + ⑤ - ④ + ... + ⑦ - ⑧

$$b_p = \begin{bmatrix} a_{1p} \\ a_{2p} \\ \vdots \\ a_{np} \end{bmatrix}$$

$$\text{原式} = (b_1 + b_n, b_1 + b_2, b_2 + b_3, \dots, b_{n-1} + b_n)$$

$$\cancel{b_1 + b_n} - \cancel{b_1} - \cancel{b_2} + \cancel{b_2} + \cancel{b_3} - \cancel{b_3} - b_4 + \dots + b_{n-2} + b_{n-1} - b_{n-1} - b_n = 0$$

$\therefore$ 此时列 1 为 0, 原式 = 0

② 若 n 为奇数,

此时列 1 为 $2 \cdot b_n$

此时 ⑧ - ① = b_{n-1} , ... 重复操作可得

原式 = $2 \cdot |b_n, b_1, \dots, b_{n-1}|$, 再换位 $n-1$ 次, 得

$$\text{原式} = 2 \cdot |A|, \text{ 其中 } A = (a_{ij})$$

$$(4) \begin{vmatrix} 1 & 2 & 3 & \cdots & n-2 & n-1 & n \\ 2 & 3 & 4 & \cdots & n-1 & n & n \\ 3 & 4 & 5 & \cdots & n & n & n \\ \vdots & \vdots & \vdots & & \vdots & \vdots & \vdots \\ n & n & n & \cdots & n & n & n \end{vmatrix}.$$

②-①, ③-②, ..., ⑦-⑥, 得

$$\begin{vmatrix} 1 & 2 & 3 & \cdots & n-2 & n-1 & n \\ 1 & 1 & 1 & \cdots & 1 & 1 & 0 \\ 1 & 1 & 1 & \cdots & 1 & 0 & 0 \\ \vdots & \vdots & \vdots & & \vdots & \vdots & \vdots \\ 1 & 0 & 0 & \cdots & 0 & 0 & 0 \end{vmatrix} = \begin{vmatrix} 0 & 0 & 0 & \cdots & 0 & 0 & n \\ 0 & 0 & 0 & \cdots & 0 & 1 & 0 \\ 0 & 0 & 0 & \cdots & 1 & 0 & 0 \\ \vdots & \vdots & \vdots & & \vdots & \vdots & \vdots \\ 0 & 1 & 0 & \cdots & 0 & 0 & 0 \\ 1 & 0 & 0 & \cdots & 0 & 0 & 0 \end{vmatrix}$$

$$= (-1)^{\frac{n(n-1)}{2}} \cdot n \quad (\text{注意逆序数})$$

$$(1) \begin{vmatrix} a & 1 & 1 & \cdots & 1 \\ 1 & a & 1 & \cdots & 1 \\ \vdots & \vdots & \vdots & & \vdots \\ 1 & 1 & 1 & \cdots & a \end{vmatrix};$$

$$(1) \begin{vmatrix} a & 1 & 1 & \cdots & 1 \\ 1-a & a-1 & 0 & \cdots & 0 \\ 0 & 1-a & a-1 & \cdots & 0 \\ \vdots & \vdots & \vdots & & \vdots \\ 0 & 0 & 0 & \cdots & a-1 \end{vmatrix} = (a-1)^{n-1} \begin{vmatrix} a & 1 & 1 & \cdots & 1 \\ -1 & 1 & 0 & \cdots & 0 \\ 0 & -1 & 1 & \cdots & 0 \\ \vdots & \vdots & \vdots & & \vdots \\ 0 & 0 & 0 & \cdots & 1 \end{vmatrix}$$

作初等列变换, 全加到列 1.

$$= \begin{vmatrix} a+n-1 & 1 & 1 & \cdots & 1 \\ 0 & 1 & 0 & \cdots & 0 \\ 0 & -1 & 1 & \cdots & 0 \\ \vdots & \vdots & \vdots & & \vdots \\ 0 & 0 & 0 & \cdots & 1 \end{vmatrix} = \begin{vmatrix} a+n-1 & a+n-2 & \cdots & 1 \\ 0 & 1 & \cdots & 0 \\ 0 & 0 & \cdots & 0 \\ \vdots & \vdots & & \vdots \\ 0 & 0 & \cdots & 1 \end{vmatrix}$$

$$= (a-1)^{n-1} [a+n-1]$$

例 7 计算下述 n 阶行列式:

$$D_n = \begin{vmatrix} a+b & ab & 0 & 0 & \cdots & 0 & 0 \\ 1 & a+b & ab & 0 & \cdots & 0 & 0 \\ 0 & 1 & a+b & ab & \cdots & 0 & 0 \\ \vdots & \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & 0 & 0 & \cdots & 1 & a+b \end{vmatrix}$$

其中 $a \neq b$.

$$D_1 = a+b, \quad D_2 = a^2 + ab + b^2$$

$$D_n = (a+b) D_{n-1} - ab \cdot \begin{vmatrix} 1 & ab & 0 & \cdots & 0 & 0 \\ 0 & a+b & ab & \cdots & 0 & 0 \\ 0 & 1 & a+b & \cdots & 0 & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & 0 & \cdots & 1 & a+b \end{vmatrix}$$

$$= (a+b) D_{n-1} - ab \cdot D_{n-2} + ab^2 \cdot \begin{vmatrix} 0 & ab & 0 & \cdots & 0 & 0 \\ 0 & a+b & ab & \cdots & 0 & 0 \\ 0 & 1 & a+b & \cdots & 0 & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & 0 & \cdots & 1 & a+b \end{vmatrix}$$

$$D_n = (a+b) D_{n-1} - ab D_{n-2}$$

$$x^2 - (a+b)x + ab = 0, \quad (x-a)(x-b) = 0, \quad x_1 = a, \quad x_2 = b$$

$$\therefore D_n - a D_{n-1} = b \cdot (D_{n-1} - a D_{n-2})$$

$$\frac{D_n - a D_{n-1}}{D_{n-1} - a D_{n-2}} = b, \quad D_2 - a D_1 = b^2$$

$$\therefore D_n - a D_{n-1} = b^n$$

$$\frac{D_n}{a^n} - \frac{D_{n-1}}{a^{n-1}} = \left(\frac{b}{a}\right)^n$$

$$\frac{D_n}{a^n} = \frac{a+b}{a} + \sum_{k=2}^n \left(\frac{b}{a}\right)^k$$

$$\therefore D_n = (a+b) \cdot a^{n-1} + \sum_{k=2}^n b^k \cdot a^{n-k}$$

另解: $D_n - a D_{n-1} = b^n$
 $D_n - b D_{n-1} = a^n \Rightarrow D_n = \frac{a^{n+1} - b^{n+1}}{a - b}$

例 8 计算下述 n 阶行列式:

$$\begin{vmatrix} 1 & 2 & 3 & \cdots & n-1 & n \\ n & 1 & 2 & \cdots & n-2 & n-1 \\ n-1 & n & 1 & \cdots & n-3 & n-2 \\ \vdots & \vdots & \vdots & & \vdots & \vdots \\ 2 & 3 & 4 & \cdots & n & 1 \end{vmatrix}.$$

每一行的和相等

$$\frac{n(n+1)}{2} \begin{vmatrix} 1 & 2 & 3 & \cdots & n-1 & n \\ 1 & 1 & 2 & \cdots & n-2 & n-1 \\ 1 & n & 1 & \cdots & n-3 & n-2 \\ \vdots & \vdots & \vdots & & \vdots & \vdots \\ 1 & 3 & 4 & \cdots & n & 1 \end{vmatrix} = \frac{n(n+1)}{2} \begin{vmatrix} 1 & 2 & 3 & \cdots & n-1 & n \\ 0 & -1 & -1 & \cdots & -1 & -1 \\ 0 & n-1 & -1 & \cdots & -1 & -1 \\ \vdots & \vdots & \vdots & & \vdots & \vdots \\ 0 & -1 & -1 & \cdots & n-1 & -1 \end{vmatrix}$$

$$= \frac{n(n+1)}{2} \begin{vmatrix} -1 & -1 & \cdots & -1 & -1 \\ n-1 & -1 & \cdots & -1 & -1 \\ \vdots & \vdots & & \vdots & \vdots \\ -1 & -1 & \cdots & n-1 & -1 \end{vmatrix} = \frac{n(n+1)}{2} \begin{vmatrix} -1 & -1 & \cdots & -1 & -1 \\ n & 0 & \cdots & 0 & 0 \\ 0 & n & \cdots & 0 & 0 \\ \vdots & \vdots & & \vdots & \vdots \\ 0 & 0 & \cdots & n & 0 \end{vmatrix} \quad (n-1 \text{ 阶})$$

$$= \frac{n(n+1)}{2} (-n) \begin{vmatrix} -1 & -1 & \cdots & -1 & -1 \\ n & 0 & \cdots & 0 & 0 \\ 0 & n & \cdots & 0 & 0 \\ \vdots & \vdots & & \vdots & \vdots \\ 0 & 0 & \cdots & n & 0 \end{vmatrix} \quad \text{形成递推}$$

$$\text{原式} = \frac{(n+1)}{2} \cdot (-1)^{n-1} \cdot n^{n-1}$$

$$\begin{bmatrix} 1 & 1 & 1 & 1 & -7 \\ 1 & 0 & 3 & -1 & 8 \\ 1 & 2 & -1 & 1 & 2a+2 \\ 3 & 3 & 3 & 2 & -11 \\ 2 & 2 & 2 & 1 & 2a \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 1 & 1 & 1 & -7 \\ 0 & -1 & 2 & -2 & 15 \\ 0 & 1 & -2 & 0 & 2a+9 \\ 0 & 0 & 0 & -1 & 10 \\ 0 & 0 & 0 & -1 & 2a+14 \end{bmatrix}$$

$$\therefore 2a+14=10, \quad a=-2.$$

① $a \neq -2$, 无解

$$\textcircled{2} \quad a=-2, \quad \begin{bmatrix} 1 & 1 & 1 & 0 & 3 \\ 0 & 1 & -2 & 0 & 5 \\ 0 & 0 & 0 & 1 & -10 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & 1 & 0 & 3 \\ 0 & 1 & -2 & 0 & 5 \\ 0 & 0 & 0 & 1 & -10 \end{bmatrix} \quad \begin{cases} x_1 = 3 - x_3 \\ x_2 = 5 + 2x_3 \\ x_4 = -10 \end{cases}$$

$$\begin{bmatrix} 1 & -4 & -2 & -1 \\ 1 & -1 & 4 & 2 \\ 5 & -5 & -4 & 1 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & -4 & -2 & -1 \\ 0 & 3 & 6 & 3 \\ 0 & 15 & 6 & 6 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & -4 & -2 & -1 \\ 0 & 1 & 2 & 1 \\ 0 & 0 & -8 & -3 \end{bmatrix}$$

$$\begin{cases} x_3 = \frac{3}{8} \\ x_2 = \frac{1}{4} \\ x_1 = \frac{3}{4} \end{cases}$$

· 双变量

$$\begin{bmatrix} 1 & -3 & -5 \\ 2 & -1 & -4 \\ 4 & -9 & a \\ 5 & b & -35 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & -3 & -5 \\ 0 & -1 & 6 \\ 0 & 3 & a+20 \\ 0 & b+15 & -30 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & -3 & -5 \\ 0 & 1 & -6 \\ 0 & 0 & a+38 \\ 0 & 0 & 6b+60 \end{bmatrix}$$

$$\textcircled{1} \quad a+38=6b+60, \quad a=6b+22 \quad \text{时}$$

$$x_1 = x_2 = x_3 = 0$$

$$\textcircled{2} \quad a+38=0, \quad a=-38 \text{ 时, 有唯一解}$$

$$\textcircled{3} \quad b=-10 \text{ 时同理}$$

$$\textcircled{4} \quad a \neq 6b+22 \text{ 时, 无解}$$

$$\textcircled{5} \quad a=-38 \text{ 且 } b=-10 \text{ 时,}$$

$$\begin{cases} x_1 = 23x_3 \\ x_2 = 6x_3 \end{cases}$$

2.4

例 10 计算下述 n 阶行列式:

$$D_n = \begin{vmatrix} 1 & 1 & 1 & \cdots & 1 & 1 \\ 1 & C_2^1 & C_3^1 & \cdots & C_{n-1}^1 & C_n^1 \\ 1 & C_3^2 & C_4^2 & \cdots & C_n^2 & C_{n+1}^2 \\ \vdots & \vdots & \vdots & & \vdots & \vdots \\ 1 & C_{n-1}^{n-2} & C_n^{n-2} & \cdots & C_{2n-4}^{n-2} & C_{2n-3}^{n-2} \\ 1 & C_n^{n-1} & C_{n+1}^{n-1} & \cdots & C_{2n-3}^{n-1} & C_{2n-2}^{n-1} \end{vmatrix}.$$

$$C_n^k = C_{n-1}^{k-1} + C_{n-1}^k$$

$$\begin{cases} C_n^k - C_{n-1}^k = C_{n-1}^{k-1} \\ C_n^k - C_{n-1}^{k-1} = C_{n-1}^k \end{cases}$$

 $\therefore D_n$ 临行相减:

$$\begin{vmatrix} 1 & 1 & 1 & \cdots & 1 & 1 \\ 0 & 1 & 2 & \cdots & n-2 & n-1 \\ 0 & C_2^2 & C_3^2 & \cdots & C_{n-1}^2 & C_n^2 \\ \vdots & \vdots & \vdots & & \vdots & \vdots \\ 0 & C_{n-2}^{n-2} & C_{n-1}^{n-2} & \cdots & C_{2n-5}^{n-2} & C_{2n-4}^{n-2} \\ 0 & C_{n-1}^{n-1} & C_n^{n-1} & \cdots & C_{2n-4}^{n-1} & C_{2n-3}^{n-1} \end{vmatrix} = \begin{vmatrix} 1 & 1 & 1 & \cdots & 1 & 1 \\ 1 & C_2^1 & C_3^1 & \cdots & C_{n-2}^1 & C_{n-1}^1 \\ 1 & C_3^2 & C_4^2 & \cdots & C_{n-1}^2 & C_n^2 \\ \vdots & \vdots & \vdots & & \vdots & \vdots \\ 1 & C_{n-1}^{n-1} & C_n^{n-1} & \cdots & C_{2n-5}^{n-1} & C_{2n-4}^{n-1} \\ 1 & C_n^{n-1} & C_{n+1}^{n-1} & \cdots & C_{2n-4}^{n-1} & C_{2n-3}^{n-1} \end{vmatrix}$$

$$\therefore D_n = D_{n-1} = \cdots = 1$$

2. 计算下述 n 阶行列式 ($n \geq 2$):

$$D_n = \begin{vmatrix} a_1 & a_2 & a_3 & \cdots & a_{n-1} & a_n \\ 1 & -1 & 0 & \cdots & 0 & 0 \\ 0 & 2 & -2 & \cdots & 0 & 0 \\ \vdots & \vdots & \vdots & & \vdots & \vdots \\ 0 & 0 & 0 & \cdots & n-1 & 1-n \end{vmatrix}.$$

~~原式 = $(-1)^{n+1} \cdot a_n \cdot (n-1)! + (1-n) \cdot D_{n-1}$~~

全加到第一列, 原式 =

$$\begin{vmatrix} \sum_{i=1}^n a_i & a_2 & \cdots & a_{n-1} & a_n \\ 0 & -1 & & & \\ 0 & 2 & & & \\ \vdots & & & & \\ 0 & & & & \end{vmatrix} = (-1)^{n-1} \cdot (n-1)! \left(\sum_{i=1}^n a_i \right)$$

4. 计算下述 n 阶行列式:

$$D_n = \begin{vmatrix} 2a & a^2 & 0 & 0 & \cdots & 0 & 0 & 0 \\ 1 & 2a & a^2 & 0 & \cdots & 0 & 0 & 0 \\ 0 & 1 & 2a & a^2 & \cdots & 0 & 0 & 0 \\ \vdots & \vdots & \vdots & \vdots & & \vdots & \vdots & \vdots \\ 0 & 0 & 0 & 0 & \cdots & 1 & 2a & a^2 \\ 0 & 0 & 0 & 0 & \cdots & 0 & 1 & 2a \end{vmatrix}.$$

$$D_n = 2a \cdot D_{n-1} - \begin{vmatrix} a^2 & 0 & 0 & \cdots & 0 & 0 & 0 \\ 1 & 2a & a^2 & \cdots & 0 & 0 & 0 \\ 0 & 1 & 2a & \cdots & 0 & 0 & 0 \\ \vdots & \vdots & \vdots & & \vdots & \vdots & \vdots \\ 0 & 0 & 0 & \cdots & 1 & 2a & a^2 \\ 0 & 0 & 0 & \cdots & 0 & 1 & 2a \end{vmatrix}$$

$$= 2a \cdot D_{n-1} - a^2 D_{n-2}$$

注:第4题在得到了 $D_n = 2aD_{n-1} - a^2 D_{n-2} (n \geq 3)$ 之后,利用本套书下册 8.1 节例 46 关于递推方程的解的结论立即得到

$$D_n = D_1 a^{n-1} + \frac{D_2 - aD_1}{a} (n-1)a^{n-1} = 2a^n + (n-1)a^n = (n+1)a^n (n \geq 3)$$

$n=1,2$ 时,上式也成立。

6. 计算下述 n 阶行列式($n \geq 2$):

$$\begin{vmatrix} 1 & 2 & 2 & \cdots & 2 & 2 & 2 \\ 2 & 2 & 2 & \cdots & 2 & 2 & 2 \\ 2 & 2 & 3 & \cdots & 2 & 2 & 2 \\ \vdots & \vdots & \vdots & & \vdots & \vdots & \vdots \\ 2 & 2 & 2 & \cdots & 2 & n-1 & 2 \\ 2 & 2 & 2 & \cdots & 2 & 2 & n \end{vmatrix}.$$

$$\begin{vmatrix} 1 & 2 & 2 & \cdots & 2 & 2 & 2 \\ 1 & 0 & 0 & \cdots & 0 & 0 & 0 \\ 0 & 0 & 1 & \cdots & 0 & 0 & 0 \\ \vdots & \vdots & \vdots & & \vdots & \vdots & \vdots \\ 0 & 0 & 0 & \cdots & 0 & n-3 & 0 \\ 0 & 0 & 0 & \cdots & 0 & 0 & n-2 \end{vmatrix} = - \begin{vmatrix} 2 & 2 & 2 & \cdots & 2 & 2 \\ 0 & 1 & 0 & \cdots & 0 & 0 \\ 0 & 0 & 2 & \cdots & 0 & 0 \\ \vdots & \vdots & \vdots & & \vdots & \vdots \\ 0 & 0 & 0 & \cdots & n-3 & 0 \\ 0 & 0 & 0 & \cdots & 0 & n-2 \end{vmatrix} = -2 \cdot (n-2)!$$

7. 计算下述 n 阶行列式:

$$D_n = \begin{vmatrix} x & y & y & \cdots & y & y \\ z & x & y & \cdots & y & y \\ z & z & x & \cdots & y & y \\ \vdots & \vdots & \vdots & & \vdots & \vdots \\ z & z & z & \cdots & x & y \\ z & z & z & \cdots & z & x \end{vmatrix}, y \neq z.$$

$$\begin{vmatrix} x & y & y & \cdots & y & y \\ z-x & x-y & 0 & \cdots & 0 & 0 \\ 0 & z-x & x-y & \cdots & 0 & 0 \\ \vdots & \vdots & \vdots & & \vdots & \vdots \\ 0 & 0 & 0 & \cdots & x-y & 0 \\ 0 & 0 & 0 & \cdots & z-x & x-y \end{vmatrix} = x(x-y)^{n-1} - (z-x) \cdot \begin{vmatrix} y & y & y & \cdots & y & y \\ z-x & x-y & 0 & \cdots & 0 & 0 \\ 0 & z-x & x-y & \cdots & 0 & 0 \\ \vdots & \vdots & \vdots & & \vdots & \vdots \\ 0 & 0 & 0 & \cdots & x-y & 0 \\ 0 & 0 & 0 & \cdots & z-x & x-y \end{vmatrix} \xrightarrow{D_{n-1}}$$

$$= x \cdot (x-y)^{n-1} - (z-x) \cdot y \cdot (x-y)^{n-2} + (z-x)^2 \cdot \begin{vmatrix} y & y & y & \cdots & y & y \\ z-x & x-y & 0 & \cdots & 0 & 0 \\ 0 & z-x & x-y & \cdots & 0 & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & 0 & x-y & z-x & x-y \end{vmatrix}$$

$$\therefore D_n = y \cdot (x-y)^{n-1} - (z-x) \cdot D_{n-1}, \quad D_2 = \begin{vmatrix} y & y \\ z-x & x-y \end{vmatrix} = y(z-x-y-z), \quad D_1 = y$$

①若 $x=z$, 则 $D_n = x \cdot (x-y)^{n-1}$

②若 $x=y$, 则 $D_n = (z-x) \cdot D_{n-1} = (z-x)^{n-1} y$, $\therefore$

$$\therefore D_n = x \cdot (x-y)^{n-1} - (z-x)^{n-1} y = -(z-x)^{n-1} y$$

③

通解: $D_n = (x-z) \cdot D_{n-1}$, $\therefore D_n = (x-z)^{n-1} D_1 = y \cdot (x-z)^{n-1}$

特解: 设 $D_n = k \cdot (x-y)^n$, 则:

$$\begin{aligned} k \cdot (x-y)^n &= y \cdot (x-y)^{n-1} + (x-z) \cdot k \cdot (x-y)^{n-1} \\ &= (y + k(x-z)) \cdot (x-y)^{n-1} \end{aligned}$$

$$\therefore k(x-y) = y + k(x-z)$$

$$k = \frac{y}{z-y}, \quad D_n^* = \frac{y}{z-y} \cdot (x-y)^n$$

$$\therefore D_n = y \cdot (x-z)^{n-1} + \frac{y}{z-y} \cdot (x-y)^n$$

$$D_n = x \cdot (x-y)^{n-1} - (z-x) \cdot \left[y \cdot (x-z)^{n-2} + \frac{y}{z-y} (x-y)^{n-1} \right]$$

$$= x \cdot (x-y)^{n-1} + y \cdot (x-z)^{n-1} + \frac{(x-y)^{n-1} (x-z)}{z-y}$$

$$= (x-y)^{n-1} \left(x + \frac{x-z}{z-y} \right) + y \cdot (x-z)^{n-1} \quad ?$$

$$D_n = \begin{vmatrix} x & y & y & \cdots & y & y+0 \\ z & x & y & \cdots & y & y+0 \\ z & z & x & \cdots & y & y+0 \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ z & z & z & \cdots & x & y+0 \\ z & z & z & \cdots & z & (x-y)+y \end{vmatrix} = (x-y) \cdot D_{n-1} + \begin{vmatrix} x & y & y & \cdots & y & y \\ z-x & x-y & 0 & \cdots & 0 & 0 \\ 0 & z-x & x-y & \cdots & 0 & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & 0 & x-y & 0 & 0 \\ 0 & 0 & 0 & z-x & 0 & 0 \end{vmatrix}$$

关键是配凑最后一列都是 y

$$= (x-y) \cdot D_{n-1} + (-1)^{n+1} \cdot y \cdot (z-x)^{n-1}$$

$$= (x-y) D_{n-1} + y \cdot (x-z)^{n-1}$$

$$\text{而 } D'_n = D_n = (x-z) \cdot D_{n-1} + z \cdot (x-y)^{n-1}$$

$$\therefore D_n = \frac{z(x-y)^n - y(x-z)^n}{z-y}$$

8. 计算下述 n 阶行列式 ($n \geq 2$):

$$\begin{vmatrix} 1 & 2 & 3 & \cdots & n-1 & n \\ 2 & 3 & 4 & \cdots & n & 1 \\ 3 & 4 & 5 & \cdots & 1 & 2 \\ \vdots & \vdots & \vdots & & \vdots & \vdots \\ n & 1 & 2 & \cdots & n-2 & n-1 \end{vmatrix}$$

$$\begin{vmatrix} 1 & 2 & 3 & \cdots & n-1 & n \\ 1 & 1 & 1 & \cdots & 1 & 1-n \\ 1 & 1 & 1 & \cdots & 1-n & 1 \\ \vdots & \vdots & \vdots & & \vdots & \vdots \\ 1 & 1 & 1-n & & 1 & 1 \\ 1 & 1-n & 1 & & 1 & 1 \end{vmatrix} = \begin{vmatrix} \frac{n(n+1)}{2} & 2 & 3 & \cdots & n-1 & n \\ 0 & 1 & 1 & & 1 & 1-n \\ 0 & 1 & 1 & & 1-n & 1 \\ \vdots & \vdots & \vdots & & \vdots & \vdots \\ 0 & 1 & 1-n & & 1 & 1 \\ 0 & 1-n & 1 & & 1 & 1 \end{vmatrix} = \frac{n(n+1)}{2} \begin{vmatrix} 1 & 1 & \cdots & 1 & 1-n \\ 1 & 1 & \cdots & 1-n & 1 \\ \vdots & \vdots & & \vdots & \vdots \\ 1-n & 1 & \cdots & 1 & 1 \end{vmatrix}$$

$$\frac{n(n+1)}{2} \begin{vmatrix} 1 & 2 & 3 & \cdots & n-1 & n \\ 1 & 3 & 4 & \cdots & n & 1 \\ 1 & 4 & 5 & \cdots & 1 & 2 \\ \vdots & \vdots & \vdots & & \vdots & \vdots \\ 1 & 1 & 2 & \cdots & n-2 & n-1 \end{vmatrix} = \frac{n(n+1)}{2} \begin{vmatrix} 1 & 2 & 3 & \cdots & n-1 & n \\ 0 & 1 & 1 & \cdots & 1 & 1-n \\ 0 & 1 & 1 & \cdots & 1-n & 1 \\ \vdots & \vdots & \vdots & & \vdots & \vdots \\ 0 & 1-n & 1 & \cdots & 1 & 1 \end{vmatrix}$$

$$= \frac{n(n+1)}{2} \begin{vmatrix} 1 & 1 & \cdots & 1 & 1-n \\ 1 & 1 & \cdots & 1-n & 1 \\ \vdots & \vdots & & \vdots & \vdots \\ 1-n & 1 & \cdots & 1 & 1 \end{vmatrix} = \frac{n(n+1)}{2} \begin{vmatrix} 1 & 1 & \cdots & 1 & 1-n \\ & & & & \\ & & & & \\ & & & & \\ & & & & \end{vmatrix}$$

$$(-1)^{\frac{(n-1)(n-2)}{2}+1} \cdot \frac{n+1}{2} n^{n-1} \quad ? \quad (-1)^{\frac{n(n-1)}{2}} \cdot \frac{n+1}{2} \cdot n^{n-1}$$

9. 用本节典型例题的例 9 的结果, 计算下列 n 阶行列式:

$$(1) \begin{vmatrix} 1+x_1 y_1 & 1+x_1 y_2 & \cdots & 1+x_1 y_n \\ 1+x_2 y_1 & 1+x_2 y_2 & \cdots & 1+x_2 y_n \\ \vdots & \vdots & & \vdots \\ 1+x_n y_1 & 1+x_n y_2 & \cdots & 1+x_n y_n \end{vmatrix}$$

$$a_{ij} = x_i y_j \quad \text{令 } A = (a_{ij})_{n \times n}, \text{ 则 } = |A| + \sum_{i=1}^n \sum_{j=1}^n A_{ij} = 0$$

(两列成比例值为 0)

$$(2) \begin{vmatrix} 1+t & t & t & \cdots & t \\ t & 2+t & t & \cdots & t \\ t & t & 3+t & \cdots & t \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ t & t & t & \cdots & n+t \end{vmatrix}.$$

$$n! + t \cdot (n! + \frac{n!}{2} + \frac{n!}{3} + \cdots + \frac{n!}{n})$$

$$= n! (1 + t + \frac{t}{2} + \cdots + \frac{t}{n})$$

10. 计算下述 n 阶行列式 ($n \geq 2$):

$$\begin{vmatrix} 1 & 1 & 1 & \cdots & 1 \\ 1 & a_1 & 0 & \cdots & 0 \\ 1 & 0 & a_2 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & 0 & 0 & \cdots & a_{n-1} \end{vmatrix},$$

其中 $a_1 a_2 \cdots a_{n-1} \neq 0$.

$$\text{原式} = \begin{vmatrix} 1 & 1 & 1 & \cdots & 1 \\ 1 & a_1 & 0 & \cdots & 0 \\ 1 & 0 & a_2 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & 0 & 0 & \cdots & 0 \end{vmatrix} + \begin{vmatrix} 1 & 1 & 1 & \cdots & 0 \\ 1 & a_1 & 0 & \cdots & 0 \\ 1 & 0 & a_2 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & 0 & 0 & \cdots & a_{n-1} \end{vmatrix} \quad D_n = a_1 a_2 \cdots a_{n-1} \cdot \begin{vmatrix} 1 - (\frac{1}{a_1} + \frac{1}{a_2} + \cdots + \frac{1}{a_{n-1}}) \\ 0 \\ 0 \\ \vdots \\ 0 \end{vmatrix}$$

$$= (-1)^{n+1} \begin{vmatrix} 1 & a_1 & 0 & \cdots & 0 & 0 \\ 1 & 0 & a_2 & \cdots & 0 & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ 1 & 0 & 0 & \cdots & 0 & a_{n-1} \end{vmatrix} + a_{n-1} D_{n-1} = a_1 a_2 \cdots a_{n-1} \cdot (1 - \sum_{i=1}^{n-1} \frac{1}{a_i})$$

$$= (-1)^{2n+1} \cdot a_1 a_2 \cdots a_{n-2} + a_{n-1} D_{n-1}$$

$$\therefore D_n = a_{n-1} D_{n-1} - a_1 a_2 \cdots a_{n-2}$$

11. 计算下述 n 阶行列式:

$$D_n = \begin{vmatrix} 5 & 3 & 0 & 0 & \cdots & 0 & 0 \\ 2 & 5 & 3 & 0 & \cdots & 0 & 0 \\ 0 & 2 & 5 & 3 & \cdots & 0 & 0 \\ \vdots & \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & 0 & 0 & \cdots & 2 & 5 \end{vmatrix}.$$

$$\begin{vmatrix} a_2 & a_3 & 0 & 0 & \cdots & 0 & 0 \\ a_1 & a_2 & a_3 & 0 & \cdots & 0 & 0 \\ 0 & a_1 & a_2 & a_3 & \cdots & 0 & 0 \\ \vdots & \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & 0 & 0 & \cdots & a_1 & a_2 \end{vmatrix} = a_2 D_{n-1} - a_1 \begin{vmatrix} a_3 & 0 & 0 & 0 & \cdots & 0 & 0 \\ a_1 & a_2 & a_3 & 0 & \cdots & 0 & 0 \\ 0 & a_1 & a_2 & a_3 & \cdots & 0 & 0 \\ \vdots & \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & 0 & 0 & \cdots & a_1 & a_2 \end{vmatrix}$$

$$= a_2 D_{n-1} - a_1 a_3 D_{n-2}.$$

$$\therefore D_n = a_2 D_{n-1} - a_1 a_3 D_{n-2},$$

$$x^2 - a_2 x + a_1 a_3 = 0, \Delta = a_2^2 - 4a_1 a_3.$$

$$\textcircled{1} \text{ 若 } a_2^2 - 4a_1 a_3 \geq 0,$$

$$x = \frac{a_2 \pm \sqrt{a_2^2 - 4a_1 a_3}}{2}$$

$$D_n - x_1 D_{n-1} = x_2 (D_{n-1} - x_1 D_{n-2}), \text{ 而 } D_2 = a_2^2 - a_1 a_3, D_1 = a_2.$$

$$D_n - x_1 D_{n-1} = x_2^{n-1} (D_2 - x_1 D_1)$$

....

$$\textcircled{2} a_2^2 - 4a_1 a_3 < 0, \text{ 有两复数根, } a_n = (e^{\alpha n} (C_1 \cos(\beta n) + C_2 \sin(\beta n))) \dots$$

$$\Delta = 25 - 24 = 1, \quad x = \frac{5 \pm 1}{2} = 3/2$$

$$D_n - 2D_{n-1} = 3^{n-1} (D_2 - 2D_1) = 9 \cdot 3^{n-1} = 3^{n+1}$$

$$D_n - 3D_{n-1} = 2^{n-1} (D_2 - 3D_1) = 2^{n+1}$$

$$\therefore D_n = 3^{n+1} - 2^{n+1}$$

12. 计算下述 n 阶行列式:

$$D_n = \begin{vmatrix} 1+x^2 & x & 0 & 0 & \cdots & 0 & 0 \\ x & 1+x^2 & x & 0 & \cdots & 0 & 0 \\ 0 & x & 1+x^2 & x & \cdots & 0 & 0 \\ \vdots & \vdots & \vdots & \vdots & & \vdots & \vdots \\ 0 & 0 & 0 & 0 & \cdots & x & 1+x^2 \end{vmatrix}.$$

$$D_n = (x^2+1) \cdot D_{n-1} - x \cdot \begin{vmatrix} x & & \\ x & 1+x^2 & x \end{vmatrix}$$

$$= (x^2+1) \cdot D_{n-1} - x^2 \cdot D_{n-2}$$

$$\therefore D_n - x^2 D_{n-1} = D_{n-1} - x^2 D_{n-2} = D_2 - x^2 D_1 = x^4 + x^2 + 1 - x^4 - x^2 = 1$$

$$\therefore D_n = x^2 \cdot D_{n-1} + 1$$

$$D_n - D_{n-1} = x^2 (D_{n-1} - D_{n-2}) = x^{2n-4} \cdot x^4 = x^{2n}$$

$$\therefore D_n = D_{n-1} + x^{2n}$$

$$\therefore (x^2-1) \cdot D_{n-1} = x^{2n} - 1 = (x^2-1) \cdot (1+x^2+\cdots+x^{2n-2})$$

$$\therefore D_n = 1+x^2+x^4+\cdots+x^{2n}$$

13. 计算下述 n 阶行列式 ($n \geq 2$):

$$\begin{vmatrix} 1 & 1 & \cdots & 1 \\ x_1 + 1 & x_2 + 1 & \cdots & x_n + 1 \\ x_1^2 + x_1 & x_2^2 + x_2 & \cdots & x_n^2 + x_n \\ x_1^3 + x_1^2 & x_2^3 + x_2^2 & \cdots & x_n^3 + x_n^2 \\ \vdots & \vdots & \ddots & \vdots \\ x_1^{n-1} + x_1^{n-2} & x_2^{n-1} + x_2^{n-2} & \cdots & x_n^{n-1} + x_n^{n-2} \end{vmatrix}$$

按行进行拆分!

$$\begin{vmatrix} 1 & 1 & \cdots & 1 \\ x_1 & x_2 & \cdots & x_n \\ x_1^2 - 1 & x_2^2 - 1 & \cdots & x_n^2 - 1 \\ x_1^3 - x_1 & x_2^3 - x_2 & \cdots & x_n^3 - x_n \\ \vdots & \vdots & \ddots & \vdots \\ x_1^{n-1} - x_1^{n-2} & x_2^{n-1} - x_2^{n-2} & \cdots & x_n^{n-1} - x_n^{n-2} \end{vmatrix} = \begin{vmatrix} 1 & 1 & \cdots & 1 \\ x_1 & x_2 & \cdots & x_n \\ x_1^2 & x_2^2 & \cdots & x_n^2 \\ x_1^3 & x_2^3 & \cdots & x_n^3 \\ \vdots & \vdots & \ddots & \vdots \\ x_1^{n-1} & x_2^{n-1} & \cdots & x_n^{n-1} \end{vmatrix}$$

$$= \prod_{1 \leq i < j \leq n} (x_i - x_j)$$

* 14. 计算下述 n 阶行列式:

$$D_n = \begin{vmatrix} 1 - a_1 & a_2 & 0 & 0 & \cdots & 0 & 0 \\ -1 & 1 - a_2 & a_3 & 0 & \cdots & 0 & 0 \\ 0 & -1 & 1 - a_3 & a_4 & \cdots & 0 & 0 \\ \vdots & \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & 0 & 0 & \cdots & -1 & 1 - a_n \end{vmatrix}$$

$$\begin{vmatrix} a_2 - a_1 & a_2 & 0 & 0 & \cdots & 0 & 0 \\ a_2 - a_2 & 1 - a_2 & a_3 & 0 & \cdots & 0 & 0 \\ a_4 - a_3 & -1 & 1 - a_3 & a_4 & \cdots & 0 & 0 \\ \vdots & \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ -a_n & 0 & 0 & 0 & \cdots & -1 & 1 - a_n \end{vmatrix}$$

(2~n全加第一行, 再3~n全加第二行...)

$$\begin{vmatrix} -a_1 & 0 & 0 & 0 & \cdots & 0 & 1 \\ -1 & 1 - a_2 & a_3 & 0 & \cdots & 0 & 0 \\ 0 & -1 & 1 - a_3 & a_4 & \cdots & 0 & 0 \\ \vdots & \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & 0 & 0 & \cdots & -1 & 1 - a_n \end{vmatrix} = \begin{vmatrix} -a_1 & 0 & 0 & 0 & \cdots & 0 & 1 \\ -1 & -a_2 & 0 & 0 & \cdots & 0 & 1 \\ 0 & -1 & -a_3 & 0 & \cdots & 0 & 1 \\ \vdots & \vdots & \vdots & \vdots & \ddots & \vdots & 1 \\ 0 & 0 & 0 & 0 & \cdots & -a_{n-1} & 1 \\ 0 & 0 & 0 & 0 & \cdots & -1 & 1 - a_n \end{vmatrix}$$

$$= (-1)^{n-1} a_1 a_2 \cdots a_{n-1} (-1 - a_n) + \begin{vmatrix} -a_1 & \cdots & 0 & 1 \\ -1 & -a_2 & \cdots & 0 & 1 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & \cdots & -a_{n-2} & 1 \\ 0 & 0 & \cdots & -1 & 1 \end{vmatrix} \rightarrow D_{n-1}$$

$$\begin{aligned}
 D_2 &= \begin{vmatrix} -a_1 & 1 \\ -1 & 1 \end{vmatrix} = 1 - a_1 \\
 &= (-1)^{n+1} \cdot a_1 a_2 \cdots a_{n-1} + (-1)^n \cdot a_1 a_2 \cdots a_n + (-1)^{n-2} \cdot a_1 a_2 \cdots a_{n-2} + D_{n-2} \\
 &= \sum_{i=1}^n \left[(-1)^i \cdot \prod_{j=1}^i a_j \right] + 1
 \end{aligned}$$

15. 计算下述 $n+1$ 阶行列式:

$$D_{n+1} = \begin{vmatrix} 1 & 0 & 0 & 0 & \cdots & 0 & 1 \\ 1 & a_1 & 0 & 0 & \cdots & 0 & 0 \\ 1 & 1 & a_2 & 0 & \cdots & 0 & 0 \\ 1 & 0 & 1 & a_3 & \cdots & 0 & 0 \\ \vdots & \vdots & \vdots & \vdots & & \vdots & \vdots \\ 1 & 0 & 0 & 0 & \cdots & 1 & a_n \end{vmatrix}$$

$$D_{n+1} = a_1 a_2 \cdots a_n + (-1)^{n+1} \begin{vmatrix} 1 & a_1 & 0 & 0 & \cdots & 0 \\ 1 & 1 & a_2 & 0 & \cdots & 0 \\ 1 & 0 & 1 & a_3 & \cdots & 0 \\ \vdots & \vdots & \vdots & \vdots & & \vdots \\ 1 & 0 & 0 & 0 & \cdots & 1 \end{vmatrix} \quad \begin{matrix} \nearrow C_n \\ \text{按最后一行展开} \end{matrix}$$

$$\begin{aligned}
 C_n &= (-1)^{n+1} a_1 a_2 \cdots a_{n-1} + C_{n-1}, \quad C_2 = 1 - a_1 \\
 \therefore C_n &= \sum_{i=1}^{n-1} \left[(-1)^i \cdot \prod_{j=1}^i a_j \right] + 1
 \end{aligned}$$

(把最后一行经过 $n-1$ 次
逐行转换到第一行)

$$D_{n+1} = -D_n + a_1 a_2 \cdots a_n$$

$$\therefore D_{n+1} = a_1 a_2 \cdots a_n - a_1 a_2 \cdots a_{n-1} + \cdots + (-1)^{n+1} a_1 + (-1)^n$$

16. 计算下述 n 阶行列式:

$$(1) B_n = \begin{vmatrix} 3 & 2 & 0 & \cdots & 0 & 0 \\ 1 & 3 & 2 & \cdots & 0 & 0 \\ \vdots & \vdots & \vdots & & \vdots & \vdots \\ 0 & 0 & 0 & \cdots & 1 & 3 \end{vmatrix};$$

(1) 典

$$(2) D_n = \begin{vmatrix} 5 & 6 & 0 & 0 & 0 & \cdots & 0 & 0 \\ 4 & 5 & 2 & 0 & 0 & \cdots & 0 & 0 \\ 0 & 1 & 3 & 2 & 0 & \cdots & 0 & 0 \\ 0 & 0 & 1 & 3 & 2 & \cdots & 0 & 0 \\ \vdots & \vdots & \vdots & \vdots & \vdots & & \vdots & \vdots \\ 0 & 0 & 0 & 0 & 0 & \cdots & 3 & 2 \\ 0 & 0 & 0 & 0 & 0 & \cdots & 1 & 3 \end{vmatrix}$$

$$(2) \quad 3 \cdot D_{n-1} - 2 \cdot \begin{vmatrix} 5 & 6 & & & & & & \\ 4 & 5 & 2 & & & & & \\ & 1 & 3 & 2 & & & & \\ & & 1 & 3 & 2 & & & \\ \vdots & \vdots & \vdots & \vdots & \vdots & & \vdots & \vdots \\ & & & & & 3 & 2 \\ & & & & & 0 & 1 \end{vmatrix} = 3D_{n-1} - 2D_{n-2}$$

$$D_n - D_{n-1} = 2 \cdot (D_{n-1} - D_{n-2}) = 2^{n-2} \cdot (D_2 - D_1) = 2^{n-3} \cdot (D_3 - D_2) = 2^{n-2} \cdot (D_2 - D_1) + 5$$

$$D_3 = 3D_2 - 2 \cdot \begin{vmatrix} 5 & 6 \\ 0 & 1 \end{vmatrix} = 3D_2 - D_1 = -2^n$$

$$\therefore D_n = 5 - (2^n + 2^{n-1} + \dots + 2^2) = -\frac{4 \cdot (1 - 2^{n-1})}{-1} = 9 - 2^{n+1}$$

17. 计算下述 n 阶行列式:

$$D_n = \begin{vmatrix} 1+x_1 & 1+x_2 & \cdots & 1+x_n \\ 1+x_1^2 & 1+x_2^2 & \cdots & 1+x_n^2 \\ \vdots & \vdots & \ddots & \vdots \\ 1+x_1^n & 1+x_2^n & \cdots & 1+x_n^n \end{vmatrix}$$

$$D_n = \begin{vmatrix} 1 & 1 & 1 & \cdots & 1 \\ 0 & 1+x_1 & 1+x_2 & \cdots & 1+x_n \\ 0 & 1+x_1^2 & 1+x_2^2 & \cdots & 1+x_n^2 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 1+x_1^n & 1+x_2^n & \cdots & 1+x_n^n \end{vmatrix} = \begin{vmatrix} 1 & 1 & 1 & \cdots & 1 \\ -1 & x_1 & x_2 & \cdots & x_n \\ -1 & x_1^2 & x_2^2 & \cdots & x_n^2 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ -1 & x_1^n & x_2^n & \cdots & x_n^n \end{vmatrix}$$

$$= \begin{vmatrix} 2 & 1 & 1 & \cdots & 1 \\ 0 & x_1 & x_2 & \cdots & x_n \\ 0 & x_1^2 & x_2^2 & \cdots & x_n^2 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & x_1^n & x_2^n & \cdots & x_n^n \end{vmatrix} - \begin{vmatrix} 1 & 1 & 1 & \cdots & 1 \\ 1 & x_1 & x_2 & \cdots & x_n \\ 1 & x_1^2 & x_2^2 & \cdots & x_n^2 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & x_1^n & x_2^n & \cdots & x_n^n \end{vmatrix}$$

$$= 2 \cdot \frac{(x_1 x_2 \cdots x_n)}{\prod_{1 \leq j < l \leq n} (x_j - x_l)} - (x_1 - 1)(x_2 - 1) \cdots (x_n - 1) \prod_{1 \leq j < l \leq n} (x_j - x_l)$$

18. 证明: 如果 n 阶行列式的某一行(或某一列)的所有元素都等于 1, 那么这个行列式的所有元素的代数余子式的和等于行列式的值。

由例 9 结论, $|A(-1)| = |A| - \sum_{i=1}^n \sum_{j=1}^n A_{ij}$, 而 $A(-1)$ 有零行

$$\therefore |A(-1)| = 0$$

Q.E.D.

19. 证明: 如果把 n 级矩阵 A 的所有元素都加上同一个数 t 得到矩阵 $A(t)$, 那么 $A(t)$ 的所有元素的代数余子式的和等于 A 的所有元素的代数余子式的和。

$$|A(t)| = |A| + t \cdot \sum_{i=1}^n \sum_{j=1}^n A_{ij}$$

$$|A(t)(-t)| = |A| = |A(t)| - t \cdot \sum_{i=1}^n \sum_{j=1}^n A(t)_{ij}$$

联立即证

3. 当 λ 取什么值时, 下述齐次线性方程组有非零解?

$$\begin{cases} (\lambda-2)x_1 - 3x_2 - 2x_3 = 0, \\ -x_1 + (\lambda-8)x_2 - 2x_3 = 0, \\ 2x_1 + 14x_2 + (\lambda+3)x_3 = 0. \end{cases}$$

4. 当 a 和 b 取什么值时, 下述齐次线性方程组有非零解?

$$\begin{cases} ax_1 + x_2 + x_3 = 0, \\ x_1 + bx_2 + x_3 = 0, \\ x_1 + 2bx_2 + x_3 = 0. \end{cases}$$

5. 当 a 和 b 取什么值时, 下述线性方程组有唯一解? 有无穷多个解? 无解?

$$\begin{cases} ax_1 + x_2 + x_3 = 2, \\ x_1 + bx_2 + x_3 = 1, \\ x_1 + 2bx_2 + x_3 = 2. \end{cases}$$

6. 下述线性方程组何时有一解? 有无穷多个解? 无解?

$$\begin{cases} ax_1 + x_2 + x_3 = 2, \\ x_1 + bx_2 + x_3 = 1, \\ x_1 + 2bx_2 + x_3 = 1. \end{cases}$$

$$3, \begin{vmatrix} \lambda-2 & -3 & -2 \\ -1 & \lambda-8 & -2 \\ 2 & 14 & \lambda+3 \end{vmatrix} = \begin{vmatrix} \lambda-2 & -3 & -2 \\ -1 & \lambda-8 & -2 \\ 0 & 2\lambda-2 & \lambda-1 \end{vmatrix}$$

$$= (\lambda-1) \begin{vmatrix} -1 & \lambda-8 & -2 \\ 0 & 2 & 1 \\ \lambda-2 & -3 & -2 \end{vmatrix} = (\lambda-1) \begin{vmatrix} -1 & \lambda-4 & 0 \\ 0 & 2 & 1 \\ \lambda-2 & 1 & 0 \end{vmatrix}$$

$$= (1-\lambda) \begin{vmatrix} -1 & \lambda-4 \\ \lambda-2 & 1 \end{vmatrix}$$

$$= (1-\lambda) \cdot (-\lambda^2 + 6\lambda - 9) = (\lambda-1) \cdot (\lambda-3)^2.$$

$\therefore \lambda=1/3$ 时有非零解

$$4, \begin{vmatrix} a & 1 & 1 \\ 1 & b & 1 \\ 1 & 2b & 1 \end{vmatrix} = \begin{vmatrix} a & 1 & 1 \\ 1 & 0 & 1 \\ 0 & b & 0 \end{vmatrix} = b \cdot \begin{vmatrix} a & 0 & 1 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{vmatrix} = -b \cdot (a-1)$$

$$\therefore b=0/a=1$$

$$5, \begin{bmatrix} a & 1 & 1 & 2 \\ 1 & b & 1 & 1 \\ 1 & 2b & 1 & 2 \end{bmatrix} \rightarrow \begin{bmatrix} a & 1 & 1 & 2 \\ 1 & b & 1 & 1 \\ 0 & b & 0 & 1 \end{bmatrix} \rightarrow \begin{bmatrix} a & 1 & 1 & 2 \\ 1 & 0 & 1 & 0 \\ 0 & b & 0 & 1 \end{bmatrix}$$

$$\rightarrow \begin{bmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 1-a & 2 \\ 0 & b & 0 & 1 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 1-a & 2 \\ 0 & 0 & b(a-1) & 1-2b \end{bmatrix}$$

$$\begin{cases} b \neq 0 \text{ 且 } a \neq 1 \text{ 时有唯一解} \\ b = 0 \text{ 时无解} \\ a = 1 \text{ 时 } \begin{cases} b \neq \frac{1}{2}, \text{ 无解} \\ b = \frac{1}{2}, \text{ 无穷多解} \end{cases} \end{cases}$$

6.

1. 计算下述行列式：

$$\begin{vmatrix} 2 & 3 & 0 & 0 & 0 \\ -1 & 4 & 0 & 0 & 0 \\ 37 & 85 & 1 & 2 & 0 \\ 29 & 73 & 0 & 3 & 4 \\ 19 & 67 & 1 & 0 & 2 \end{vmatrix}.$$

$$= 11 \cdot \begin{vmatrix} 1 & 2 & 0 \\ 0 & 3 & 4 \\ 1 & 0 & 2 \end{vmatrix} = 11 \cdot \begin{vmatrix} 1 & 0 & 2 \\ 0 & 2 & -2 \\ 0 & 3 & 4 \end{vmatrix} = 11 (14) = 154$$

2. 计算下述行列式：

$$\begin{vmatrix} a_{11} & \cdots & a_{1k} & c_{11} & \cdots & c_{1r} \\ \vdots & & \vdots & \vdots & & \vdots \\ a_{k1} & \cdots & a_{kk} & c_{k1} & \cdots & c_{kr} \\ 0 & \cdots & 0 & b_{11} & \cdots & b_{1r} \\ \vdots & & \vdots & \vdots & & \vdots \\ 0 & \cdots & 0 & b_{r1} & \cdots & b_{rr} \end{vmatrix}.$$

3. 设 $|A|$ 是关于 $1, 2, \dots, n$ 的范德蒙行列式, 计算：

(1) $A \begin{pmatrix} 1, 2, \dots, n-1 \\ 2, 3, \dots, n \end{pmatrix};$

(2) $A \begin{pmatrix} 1, 2, \dots, n-1 \\ 1, 3, \dots, n \end{pmatrix}.$

(1) $\prod_{i=1}^{n-2} i!$

(2) $(n-1) \cdot \prod_{i=1}^{n-2} i!$

补充题 2.

补充题 二

1. 在空间右手直角坐标系 $[0; \vec{e}_1, \vec{e}_2, \vec{e}_3]$ 中, 两个非零向量 $\vec{a}, \vec{b}$ 的坐标分别为 $(a_1, a_2, 0)'$, $(b_1, b_2, 0)'$ 。

(1) 求以 $\vec{a}, \vec{b}$ 为邻边的平行四边形的面积, 并且把结果用一个行列式表示;

(2) 求以 $\vec{a}, \vec{b}$ 为两边的三角形的面积, 并且把结果用一个行列式表示。

2. 在空间右手直角坐标系 $[0; \vec{e}_1, \vec{e}_2, \vec{e}_3]$ 中, 三个非零向量 $\vec{a}, \vec{b}, \vec{c}$ 的坐标分别为

$$(a_1, a_2, a_3)', (b_1, b_2, b_3)', (c_1, c_2, c_3)'.$$

求以 $\vec{a}, \vec{b}, \vec{c}$ 为棱的平行六面体的体积, 并且把结果用一个行列式表示。

$$\begin{aligned} \text{1. (1)} \quad S &= |\vec{a}| |\vec{b}| \sin \langle \vec{a}, \vec{b} \rangle = |\vec{a} \times \vec{b}| \\ \text{而 } \vec{a} \times \vec{b} &= (a_1 \vec{e}_1 + a_2 \vec{e}_2) \times (b_1 \vec{e}_1 + b_2 \vec{e}_2) \\ &= (a_1 b_2 - a_2 b_1) \vec{e}_3 \\ \therefore S &= \begin{vmatrix} a_1 & b_1 \\ a_2 & b_2 \end{vmatrix} \end{aligned}$$

(2)

$$\begin{aligned} 2, \quad V &= |\vec{a} \times \vec{b}| \cdot |\vec{c}| \cdot |\cos \langle \vec{c}, \vec{a} \times \vec{b} \rangle| \\ &= |\vec{a} \times \vec{b} \cdot \vec{c}| \end{aligned}$$

$$= \dots \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix}$$

3. 求元素为1或0的3阶行列式可取到的最大值。

4. 求元素为1或-1的3阶行列式可取到的最大值。

5. 设 $n \geq 3$, 证明: 元素为1或-1的 n 阶行列式的绝对值不超过 $(n-1)!(n-1)$ 。

6. 求元素为1或-1的4阶行列式可取到的最大值。

7. 设 $n \geq 2$, 证明: 元素为1或-1的 n 阶行列式的值能被 2^{n-1} 整除。

3, 由6项元素组成 为使取值最大, 要使正数项尽可能为1, 负号项尽可能为0

若3个正项均为1, 则全为1, 行列式为0, 则最多2个正项为1

$$\begin{vmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 1 & 0 & 1 \end{vmatrix} = 2 = \begin{vmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{vmatrix} = \begin{vmatrix} 1 & 0 & 1 \\ 1 & 1 & 0 \\ 0 & 1 & 1 \end{vmatrix} = \begin{vmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 1 & 0 & 1 \end{vmatrix}$$

$$\therefore \max = 2.$$

(正项元素位置固定)

$$4, \begin{vmatrix} 1 & 1 & 1 & 1 \\ -1 & 1 & 1 & 1 \\ -1 & -1 & 1 & 1 \end{vmatrix} = 1 + 1 - 1 - 1 + 1 + 1 = 2.$$

值必为偶数

7. 设 $n \geq 2$, 证明: 元素为 1 或 -1 的 n 阶行列式的值能被 2^{n-1} 整除。

8. 斐波那契 (Fibonacci) 数列是

$$1, 2, 3, 5, 8, 13, 21, 35, \dots$$

它满足: $F_n = F_{n-1} + F_{n-2} (n \geq 3), F_1 = 1, F_2 = 2$ 。

(1) 证明 Fibonacci 数列的通项 F_n 可由下述行列式表示:

$$F_n = \begin{vmatrix} 1 & -1 & 0 & 0 & \cdots & 0 & 0 & 0 \\ 1 & 1 & -1 & 0 & \cdots & 0 & 0 & 0 \\ 0 & 1 & 1 & -1 & \cdots & 0 & 0 & 0 \\ \vdots & \vdots & \vdots & \vdots & \ddots & \vdots & \vdots & \vdots \\ 0 & 0 & 0 & 0 & \cdots & 1 & 1 & -1 \\ 0 & 0 & 0 & 0 & \cdots & 0 & 1 & 1 \end{vmatrix};$$

(2) 求 Fibonacci 数列的通项公式。

$$8. (1) \quad F_n = F_{n-1} + \begin{vmatrix} 1 & -1 & 0 & \cdots \\ 0 & 1 & -1 & \cdots \\ \vdots & \vdots & \vdots & \ddots \end{vmatrix} = F_{n-1} + F_{n-2}, \text{ 而 } F_2 = 2, F_1 = 1.$$

(2)

9. 设 $f_{ij}(t)$ 是可微函数, $1 \leq i, j \leq n$ 。令

$$F(t) = \begin{vmatrix} f_{11}(t) & f_{12}(t) & \cdots & f_{1n}(t) \\ f_{21}(t) & f_{22}(t) & \cdots & f_{2n}(t) \\ \vdots & \vdots & \ddots & \vdots \\ f_{n1}(t) & f_{n2}(t) & \cdots & f_{nn}(t) \end{vmatrix}.$$

证明:

$$\frac{d}{dt} F(t) = \sum_{j=1}^n \begin{vmatrix} f_{11}(t) & f_{12}(t) & \cdots & \frac{d}{dt} f_{1j}(t) & \cdots & f_{1n}(t) \\ f_{21}(t) & f_{22}(t) & \cdots & \frac{d}{dt} f_{2j}(t) & \cdots & f_{2n}(t) \\ \vdots & \vdots & \ddots & \vdots & \ddots & \vdots \\ f_{n1}(t) & f_{n2}(t) & \cdots & \frac{d}{dt} f_{nj}(t) & \cdots & f_{nn}(t) \end{vmatrix}.$$

① $n=1$ 时显然成立

$$\textcircled{2} \quad n=2 \text{ 时, } F(t) = f_{11}(t) f_{22}(t) - f_{12}(t) f_{21}(t)$$

$$\begin{aligned} F'(t) &= f'_{11}(t) f_{22}(t) + f_{11}(t) f'_{22}(t) - f'_{12}(t) f_{21}(t) - f_{12}(t) f'_{21}(t) \\ &= \begin{vmatrix} f'_{11}(t) & f_{12}(t) \\ f'_{21}(t) & f_{22}(t) \end{vmatrix} + \begin{vmatrix} f_{11}(t) & f'_{12}(t) \\ f_{21}(t) & f'_{22}(t) \end{vmatrix} \end{aligned}$$

